



The parallel port of the PC is a powerful platform for implementing projects dealing with the control of real-world peripherals. It can be used to control home and other electronic appliances. The computer program through the interface circuit controls the relays, which, in turn, switch

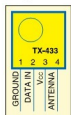


Fig. 1: Pin configuration of the RF transmitter module

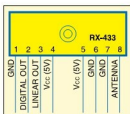


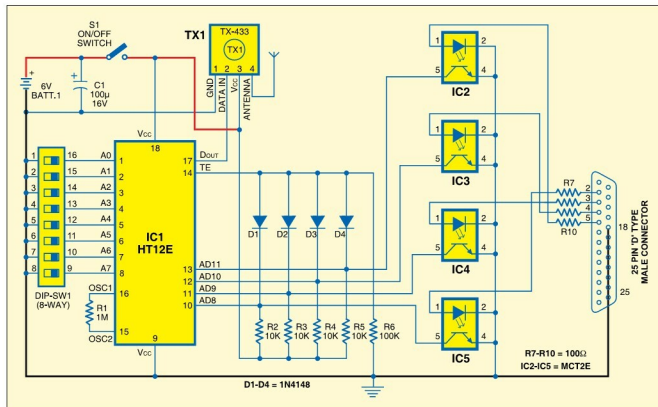
Fig. 2: Pin configuration of the RF receiver module

Here we describe how to wirelessly control electronic appliances from a remote place by using the RF module. For this PC-based wireless appliances control system, you need to design and develop the required hardware and software. The parallel port of the PC is used to control the appliances at the transmitter side. The RF interface is used instead of the IR to overcome all the drawbacks of the IR interface. The PC signals are transmitted from the RF transmitter and received by the RF receiver.

Signals from the PC's parallel port are interfaced to the RF transmitter through the RF encoder. The encoder (HT12E) continuously reads the status of the op-

V _{CC}	Output	Current
5V DC	-0 dBm	1.0 mA
12V DC	+9 dBm	3 mA

Parameter	Values
Working voltage	4.5V-5.5V DC
Bandwidth	12 MHz
Sensitivity	-103 dBm
Date rate	4800 bps
Max. data rate	9600 bps
Standby current	1.2 mA
Antenna type	Whip, strip line or helical



60 • JULY 2009 • ELECTRONICS FOR YOU